Chapter 1: Debt Growth

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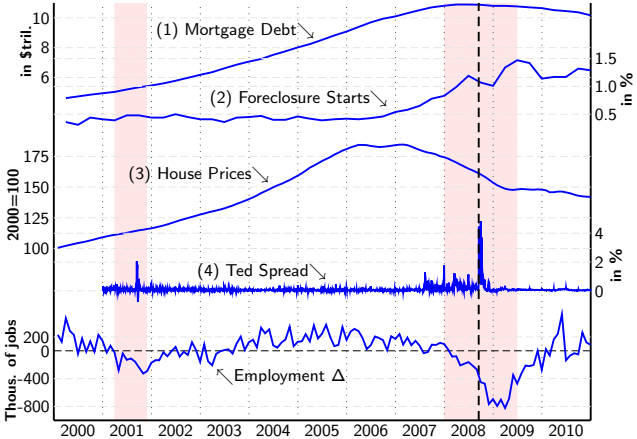
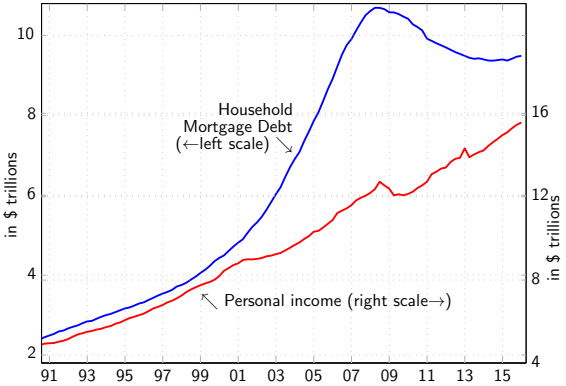
Course Overview



- (1) Debt Growth
- (2) Default
- (3) House Prices
- (4) Financial Crisis
- (5) Institutions
- (6) Expectations

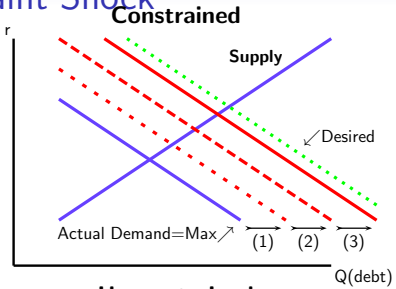
Debt Growth

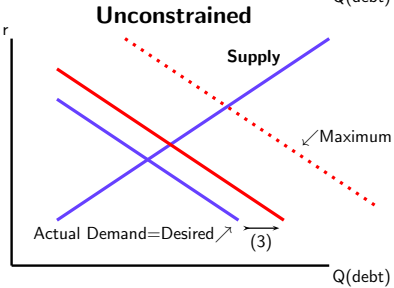


- Huge debt boom between 2001 and 2007
- Much faster than income
- Questions:
  - Where did debt grow?
  - What was the role of subprime? Securitization?
  - What was the role of the government?

(1) Exogenous Credit Constraint Shock

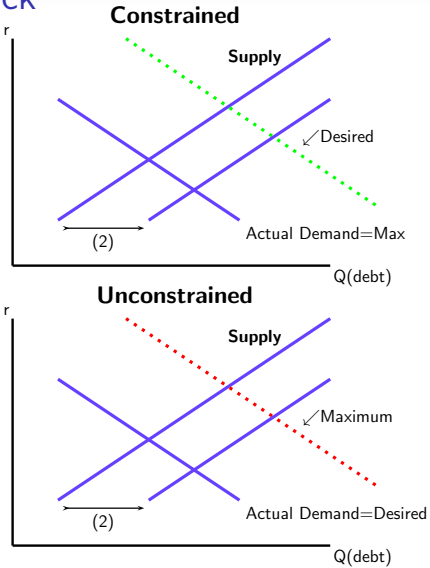
- Markets for mortgage debt: Constrained v. unconstrained
- Corbae and Quentin (2015), Favilukis et al. (2016)
- Mian and Sufi (2016): *exogenous* shock to credit constraints
  (1) *Extensive margin:* individuals not able to buy a house due to credit rationing earlier, now can.Only for constrained
  (2) *Intensive margin:* individuals with debt can now borrow more for same permanent income. Only for constrained
  (3) *House price effect:* greater credit availability boosts house prices, especially where supply more restricted. [this feeds back into intensive margin] Both, but stronger for constrained
- Credit Constraint theory implies
  - Credit constraints relaxed
  - Reallocation to constrained.





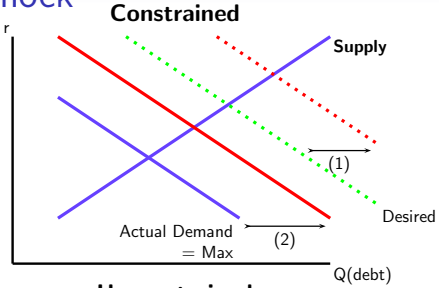
(2) Credit Supply Shock

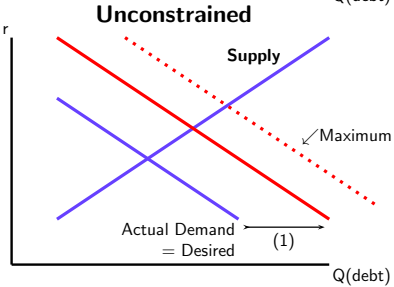
- Justiniano, Primiceri and Tambalotti (2016)
- Distinguish
  (1) "Collateral constraint"
  ...looser collateral requirements shift the demand for credit, generating counterfactual implications for interest rates and aggregate leverage
  (2) Credit Supply: "Lending constraint"
  Lending constraints are a simple modeling device to capture a combination of technological, institutional, and behavioral factors that restrain the flow of funds from savers to mortgage borrowers.
- Credit Supply shock theory implies
  - Credit constraints *not* relaxed
  - No reallocation to constrained.



(3) House Price Belief Shock

- Foote, Gerardi and Willen (2012)
  - Evolution of beliefs critical to explaining boom
    - Borrowers
    - Lenders
- Gennaioli, Shleifer and Vishny (2012)
- Kaplan, Mitman and Violante (2016) model of mortgage market
  (1) "changing beliefs about future house price growth increases demand for debt."
  (2) "increased credit flow to riskier borrowers was an endogenous response to an increased belief [among lenders] in expected house price appreciation."
- Belief shock theory implies
  - Credit constraints relaxed
  - No reallocation to constrained.





Home Mortgage Disclosure Act (HMDA)

LOAN/APPLICATION REGISTER Page ____of____

Name of Reporting Institution

City, State, Zip

Form FR HMDA-LAR

Reporter's Identification Number

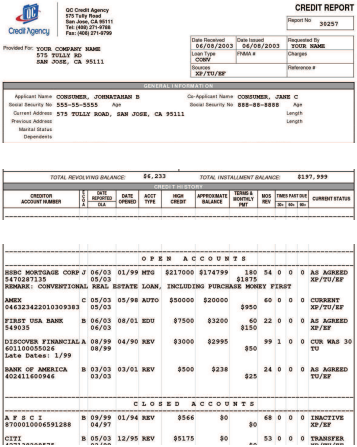
Agency Code

All columns (except Reasons for Denial) must be completed for each entry. See the instructions for details.

Application or Loan Information								Action Taken		Property Location				Applicant Information A = Applicant CA = Co-Applicant									Other Data		
														Ethnicity		Race		Sex							
Application or Loan Number	Date Application Received mm/dd/ccyy	Loan Type	Property Type	Purpose	Owner Occupancy	Loan Amount In Thousands	Preapproval	Type	Date (mm/dd/ccyy)	Five-Digit MSA/MD Number	Two-Digit State Code	Three-Digit County Code	Six-Digit Census Tract	A	CA	A	CA	A	CA	Gross Annual Income in thousands	Type of Purchaser of Loan	Reasons for Denial (optional)	Rate Spread	HOEPA Status	Lien Status

- Passed in 1975 to combat discrimination in mortgage lending
- For all loans, banks need to
  - Record application
  - Action taken
- Munnell, Alicia H., Geoffrey MB Tootell, Lynn E. Browne, and James McEneaney. "Mortgage lending in Boston: Interpreting HMDA data." *The American Economic Review* (1996): 25-53.

New York Fed Consumer Credit Panel (CCP)



GC Credit Agency
575 Tully Road
San Jose, CA 95111
Tel: (408) 271-6788
Fax: (408) 271-9799

Provided For: YOUR COMPANY NAME
575 TULLY RD
SAN JOSE, CA 95111

CREDIT REPORT

Report No: 38257

Date Received	Date Issued	Requested By
06/08/2003	06/08/2003	YOUR NAME
Loan Type: CONV	PMAN #	Charges
Sources: XP/TU/EF		Reference #

GENERAL INFORMATION

Applicant Name: CONSUMER, JOHNATHAN B
Social Security No: 555-55-5555 Age
Current Address: 575 TULLY ROAD, SAN JOSE, CA 95111
Previous Address:
Marital Status:
Dependents:

Co-Applicant Name: CONSUMER, JANE C
Social Security No: 888-88-8888 Age
Length
Length

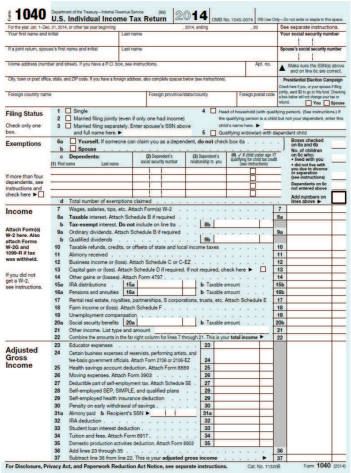
TOTAL REVOLVING BALANCE: $6,233 TOTAL INSTALLMENT BALANCE: $197,999

CREDIT HISTORY

CREDITOR ACCOUNT NUMBER	S O U R C E	DATE REPORTED DLA	DATE OPENED	ACCT TYPE	HIGH CREDIT	APPROXIMATE BALANCE	TERMS & MONTHLY PMT	MOS REV	TIMES PAST DUE 30x	60x	90x	CURRENT STATUS
OPEN ACCOUNTS												
SSRC MORTGAGE CORP J 5470287135		06/03 05/03	01/99	MTG	$217000	$174799	180 $1875	54	0	0	0	AS AGREED XP/TU/EF
REMARK: CONVENTIONAL REAL ESTATE LOAN, INCLUDING PURCHASE MONEY FIRST												
AMEX 0463234220103093B3	C	05/03 05/03	05/98	AUTO	$50000	$20000	$950	60	0	0	0	CURRENT XP/TU/EF
FIRST USA BANK 549035	B	06/03 06/03	08/01	EDU	$7500	$3200	60 $150	22	0	0	0	AS AGREED XP/EF
DISCOVER FINANCIAL 601100055026	A	08/99 08/99	04/90	REV	$3000	$2995	$50	99	1	0	0	COR WAS 30 TU
Late Dates: 1/99												
BANK OF AMERICA 402411600946	B	03/03 03/03	03/01	REV	$500	$238	$25	24	0	0	0	AS AGREED TU/EF
CLOSED ACCOUNTS												
A F S C I 870001000659128	B	09/99 04/97	01/94	REV	$566	$0	$0	68	0	0	0	INACTIVE XP/EF
CITI	B	05/03	12/95	REV	$5175	$0		53	0	0	0	TRANSFER

- 5% random sample of all individuals in Equifax every quarter.
  - 10 million individuals, dynamically updated
- By "trade line"
  - Balance
  - Delinquency status
- All debt
  - first and second liens, Home Equity Loans and Lines-of-Credit
  - Also credit cards, auto loans
- Demographic information
  - Census Block Group and age
  - No gender, race or income
- Credit Scores: Equifax™ Risk Score (not FICO™ or VantageScore™)

IRS Statistics of Income



Form 1040 Department of the Treasury—Internal Revenue Service (99) U.S. Individual Income Tax Return 2014 IRS Use Only—Do not write or staple in this space.

For the year Jan. 1–Dec. 31, 2014, or other tax year beginning , 2014, ending , 20 See separate instructions.

Your first name and initial | Last name | Your social security number

If a joint return, spouse's first name and initial | Last name | Spouse's social security number

Home address (number and street). If you have a P.O. box, see instructions. | Apt. no.

City, town or post office, state, and ZIP code. If you have a foreign address, also complete spaces below (see instructions).

Foreign country name | Foreign province/state/county | Foreign postal code

Presidential Election Campaign

Filing Status

Exemptions

Income

Adjusted Gross Income

For Disclosure, Privacy Act, and Paperwork Reduction Act Notice, see separate instructions. Cat. No. 11320B Form 1040 (2014)

- Zip code level tax return information for 1998, 2001, 2002, and 2004-present.
  - Exemptions
  - Wage and Salary
  - AGI – realized capital gains
- Problems
  - Not individual level data
  - Migration
- House price indices and data on subprime loans from Corelogic.

Survey of Consumer Finances (SCF)

- Triennial survey of individual balance sheets and income.
  - # of respondents range from 3,500 to over 5,000 per year.
  - Oversamples high income households
  - Weights offset
  - Matches aggregate data
- Contains
  - Comprehensive measures of debt
  - Income (both AGI and W&S)
  - Demographics: Age, Gender, # of Kids, Education
- No geographic information

Mian and Sufi (2009)

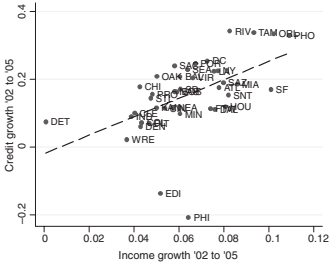
THE CONSEQUENCES OF MORTGAGE CREDIT EXPANSION: EVIDENCE FROM THE U.S. MORTGAGE DEFAULT CRISIS*

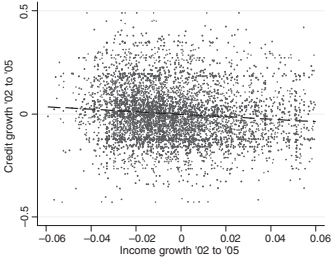
ATIF MIAN AND AMIR SUFI

We conduct a within-county analysis using detailed ZIP code–level data to document new findings regarding the origins of the biggest financial crisis since the Great Depression. The sharp increase in mortgage defaults in 2007 is significantly amplified in subprime ZIP codes, or ZIP codes with a disproportionately large share of subprime borrowers as of 1996. Prior to the default crisis, these subprime ZIP codes experience an unprecedented relative growth in mortgage credit. The expansion in mortgage credit from 2002 to 2005 to subprime ZIP codes occurs despite sharply declining relative (and in some cases absolute) income growth in these neighborhoods. In fact, 2002 to 2005 is the only period in the past eighteen years in which income and mortgage credit growth are negatively correlated. We show that the expansion in mortgage credit to subprime ZIP codes and its dissociation from income growth is closely correlated with the increase in securitization of subprime mortgages.

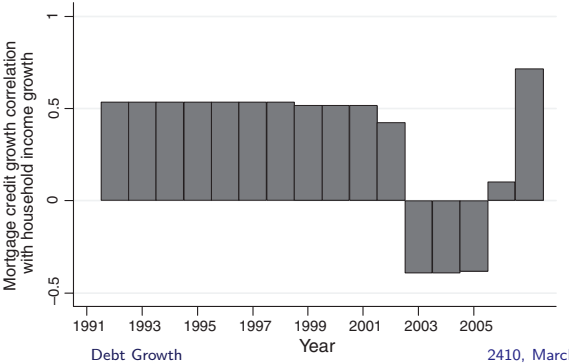
- 1393 cites on Google Scholar – one of the most cited papers about the crisis.
- Argues that "subprime" – expansion of credit to low income and low credit score borrowers caused
  1. increase in debt
  2. increase in prices
  3. foreclosures
  4. financial crisis
- Will focus on claim (1) today.

Mian and Sufi (2009)





- Correlation within versus across cities
- "In fact, 2002 to 2005 is the only period in the past eighteen years in which income and mortgage credit growth are negatively correlated. "



Regression Specification

- Regression Specification:

$$\%\Delta debt_{t,z} = \alpha_t + \alpha_z + \beta_t \%\Delta income_{t,z}$$

  where d is *total* debt t is time, z is zipcode, everything in logs
- In debt-income regressions
  - Debt = Total dollar value of all purchase mortgage originations in a zip code from HMDA
  - Income = Total income of all filers in IRS SOI.

TABLE IV
HISTORICAL MORTGAGE CREDIT GROWTH AND INCOME GROWTH CORRELATIONS

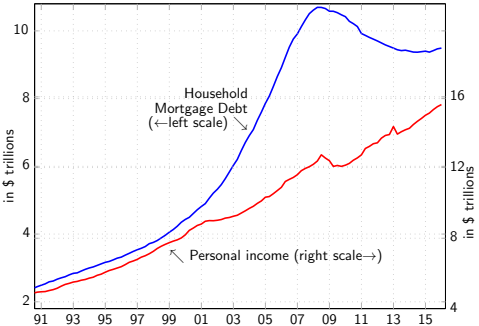
	Dependent variable: Mortgage originations for home purchase growth, annualized							
	2002–2005	1991–1998	1998–2001	2001–2002	2002–2004	2004–2005	2005–2006	2006–2007
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Income growth, annualized	−0.662**	0.537**	0.517**	0.425	−0.394**	−0.383**	0.103	0.716**
	(0.089)	(0.084)	(0.092)	(0.368)	(0.122)	(0.077)	(0.078)	(0.093)
N	3,014	2,809	3,014	3,014	3,014	3,014	3,014	3,014
R^2	.34	.55	.27	.44	.24	.39	.27	.26

Notes. This table presents the correlation across ZIP codes between mortgage origination for home purchase growth and income growth for different periods of our sample. Income growth and mortgage origination growth are measured for the exact same period for all specifications except the specification reported in column (7). We do not have income data available for 2007; as a result, in column (7) we examine the correlation between mortgage origination growth from 2006 to 2007 and income growth from 2005 to 2006. All specifications include county fixed effects.

**,* Coefficient estimate statistically distinct from 0 at the 1% and 5% levels, respectively.

Interpretation

1. "Income-based hypothesis"
   - Debt growth driven by income growth
   - Straw man



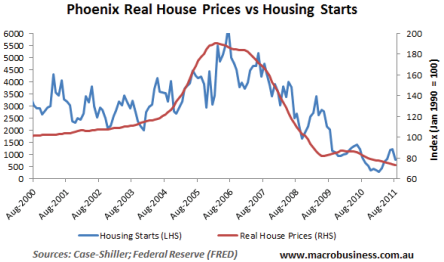
2. Credit Availability stories

2a. "Supply-based" hypotheses
   - *Exogenous* increase in credit availability
   - Institutional/legal change

2b. "Expectations-based" hypotheses
   - Change in lender beliefs about house prices

Distinguishing causes of credit expansion

- MS follow Glaeser, Gyourko and Saiz (2008)
  - Increase in demand only increases prices in places with low elasticity of supply
- MS argue that lenders should have understood this
- Expectations theory implies
  - Credit expansion only in low elasticity cities
- Find similar patterns of debt growth in high elasticity cities
- MS acknowledge that this assumes lenders understand GGS argument.
- Were lenders rational about house prices?
  - Believed 30% chance house prices would rise $> 8\%$ per year after 2005
- Was GGS a good predictor?



Credit demand based hypotheses

- Implicit assumption in MS2009 is that cross-section debt is determined by
  - income
  - credit availability
- Theories of debt demand

1 PIH implies *negative correlation between debt and income* if shocks are transitory.

2 Merton (1969)

$$\omega = \frac{\text{Investment in risky assets}}{\text{Total Wealth}} = \frac{\text{Investment in risky assets}}{\text{Financial Wealth} + \text{Human Wealth}}$$

- Eqn. (29) says *target* share of *total* wealth invested in the risky asset.

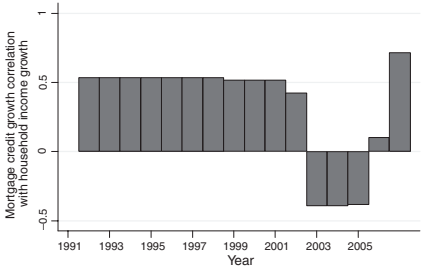
$$\omega^* = \frac{1}{RRA} \times \frac{\mathrm{E}(\tilde{R}) - R_f}{\sigma(\tilde{R})}$$

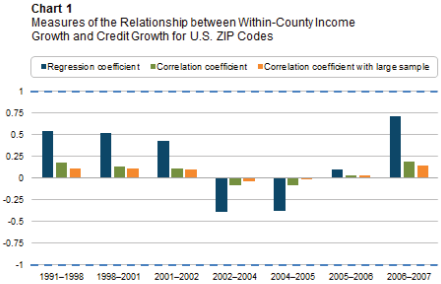
- High $\mathrm{E}(\tilde{R}) - R_f \Rightarrow$.

$$\text{Investment in risky assets} > \text{Financial Wealth}$$

- Increase in house price expectations
  - High wealth: Reduced holdings of other assets
  - Low wealth: Increased debt

Other problems





Note: The source for the regression coefficient is MS2009 (Table 4, p. 1470).

- Key figure is quite misleading

1. "Correlation" is a regression coefficient not a correlation coefficient
   - impossible to interpret
   - looks like high positive to high negative
   - but actually correlations are very low – puzzlingly
2. Do not have data for most years.
3. $\text{corr}(\%\Delta Debt_{92-93}, \%\Delta Y_{92-93}) \neq \text{corr}(\%\Delta Debt_{91-98}, \%\Delta Y_{91-98})$
4. Credit boom extended into 2007

Adelino, Schoar and Severino (2016)

1. Regress HMDA purchase originations on *HMDA* income at the individual level.
   - Find a very stable relationship

Table 3
Origination and income at the transaction level

	(1)	(2)	(3)	(4)
Ln(IRS household income)	0.600***	0.594***	0.397***	0.346***
	(0.014)	(0.018)	(0.029)	(0.039)
Ln(IRS household income)		0.038***		0.075***
× year 2004		(0.014)		(0.012)
Ln(IRS household income)		0.026		0.079***
× year 2005		(0.019)		(0.016)
Ln(IRS household income)		−0.041**		0.016
× year 2006		(0.017)		(0.016)
Year f.e. and county f.e.	Y	Y	N	N
Year f.e. and census tract f.e.	N	N	Y	Y
Number of observations	17,220,064	17,220,064	17,220,064	17,220,064
R^2	0.23	0.23	0.29	0.29

2. At zip code level regress RHS=IRS income. LHS=:
   - *average purchase amount* – very stable
   - *# of purchase loans* – not stable
   - *$ of all purchase loans* = *average purchase amount* × *# of purchase loans* – not stable

C. Panel specification

	Total purchase mortgage origination		Average mortgage size		Number of mortgages	
Ln(IRS household income)	0.378***	1.104***	0.442***	0.451***	−0.068	0.654***
	(0.080)	(0.095)	(0.033)	(0.048)	(0.073)	(0.075)
Ln(IRS household income)	−0.137***		0.010		−0.152***	
× Year 2004	(0.021)		(0.010)		(0.018)	
Ln(IRS household income)	−0.246***		0.020		−0.270***	
× Year 2005	(0.026)		(0.014)		(0.021)	
Ln(IRS household income)	−0.382***		−0.015		−0.369***	
× Year 2006	(0.027)		(0.015)		(0.023)	
ZIP code fixed effects	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y
Number of observations	36,299	36,299	36,299	36,299	36,299	36,299
R^2	0.97	0.97	0.97	0.97	0.97	0.97

Relation to Mian and Sufi I: Intensive margin

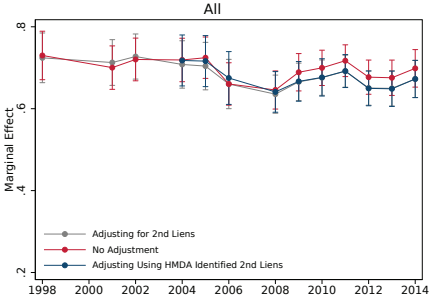
- Mian and Sufi (2015)

"As anyone who has ever obtained a mortgage knows, the income of a potential borrower almost always determines how much a banker is willing to lend. The higher the income, the more debt the borrower is allowed to carry. But..., the exact opposite was happening."

- "As any zipcode who has ever obtained a mortgage..."
- Our own version:
  - Controlling for second liens
  - With CBSA fixed effects
- Mian and Sufi (2016a) – "Fraudulent Misrepresentation" in HMDA income data. Irrelevant as this uses IRS

C. Panel specification

	Total purchase mortgage origination		Average mortgage size		Number of mortgages	
Ln(IRS household income)	0.378***	1.104***	0.442***	0.451***	−0.068	0.654***
	(0.080)	(0.095)	(0.033)	(0.048)	(0.073)	(0.075)
Ln(IRS household income) × Year 2004	−0.137***		0.010		−0.152***	
	(0.021)		(0.010)		(0.018)	
Ln(IRS household income) × Year 2005	−0.246***		0.020		−0.270***	
	(0.026)		(0.014)		(0.021)	
Ln(IRS household income) × Year 2006	−0.382***		−0.015		−0.369***	
	(0.027)		(0.015)		(0.023)	
ZIP code fixed effects	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y
Number of observations	36,299	36,299	36,299	36,299	36,299	36,299
R^2	0.97	0.97	0.97	0.97	0.97	0.97



Relation to Mian and Sufi II: Extensive margin

2 Mian and Sufi (2016b). "Extensive margin" is what matters
  - Increase in # of people getting credit
  - Increase in debt in low income areas

- Data does show decline in coefficient on income for extensive margin.

C. Panel specification

	Total purchase mortgage origination		Average mortgage size		Number of mortgages	
Ln(IRS household income)	0.378***	1.104***	0.442***	0.451***	−0.068	0.654***
	(0.080)	(0.095)	(0.033)	(0.048)	(0.073)	(0.075)
Ln(IRS household income) × Year 2004	−0.137***		0.010		−0.152***	
	(0.021)		(0.010)		(0.018)	
Ln(IRS household income) × Year 2005	−0.246***		0.020		−0.270***	
	(0.026)		(0.014)		(0.021)	
Ln(IRS household income) × Year 2006	−0.382***		−0.015		−0.369***	
	(0.027)		(0.015)		(0.023)	
ZIP code fixed effects	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y
Number of observations	36,299	36,299	36,299	36,299	36,299	36,299
R^2	0.97	0.97	0.97	0.97	0.97	0.97

(1') Correct interpretation of Mian and Sufi (2009)

- $y_2 = \beta_2 x_2$, $y_1 = \beta_1 x_1$
- Take first differences

$$y_2 - y_1 = \beta_2 y_2 - \beta_1 y_1$$

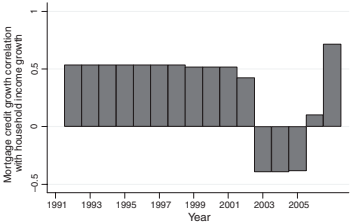
- Mian and Sufi (2009) run

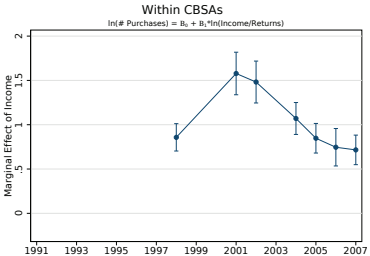
$$y_2 - y_1 = \beta_{MS}(x_2 - x_1)$$

- But what is β_{MS}?

$$\begin{aligned} y_2 - y_1 &= \beta_2(x_2 - x_1) + x_1(\beta_2 - \beta_1) \\ y_2 - y_1 &= \beta_1(x_2 - x_1) + x_2(\beta_2 - \beta_1) \end{aligned}$$

- β_{MS} only makes sense if $\beta_1 = \beta_2$
- Using our baseline panel specification



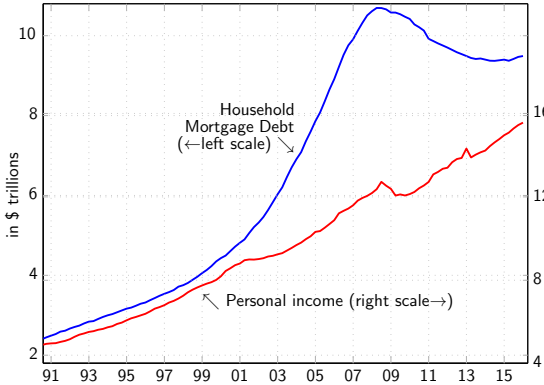


Economics of Debt

- Economic questions mostly about the stock of debt not the flow
  - MS2009, A2S about the flow
- Theories of debt demand

1. PIH implies *negative correlation between debt and income* if shocks are transitory.
2. Merton (1969)

$$\omega = \frac{\text{Investment in risky assets}}{\text{Total Wealth}} = \frac{\text{Investment in risky assets}}{\text{Financial Wealth} + \text{Human Wealth}}$$

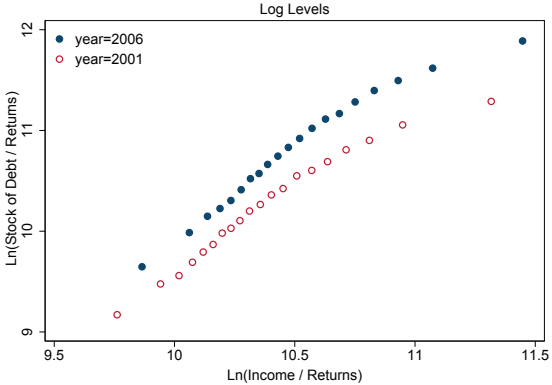


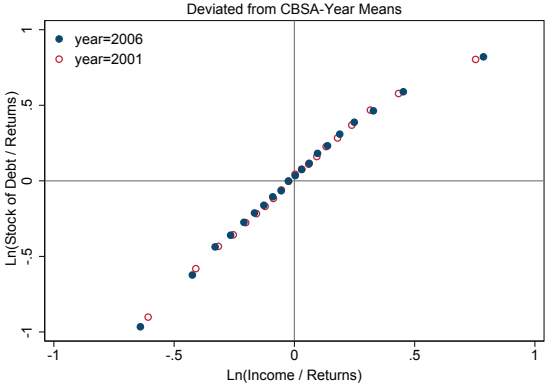
Basic Regression

$$\underbrace{y_{ict}}_{\substack{\text{• Stock } \textit{or} \text{ flow of debt} \\ \text{• Defaults} \\ \text{• Etc.}}} = \alpha_t + \underbrace{\beta_t}_{\text{Parameter of interest}} Income_{ict} + \epsilon_{ict}$$

- Panel structure
  - i indexes ZIP code (Equifax/IRS) or individual (SCF)
  - c indexes city (CBSA) – not available in SCF regressions
  - t indexes year
- Regressions are weighted by ...
  - Total tax returns in ZIP for Equifax/IRS regressions.
  - Sample weights for SCF regressions.
- Standard errors are clustered by CBSA for ZIP-level regs.

Finding 2: No relative increase in debt among low income households

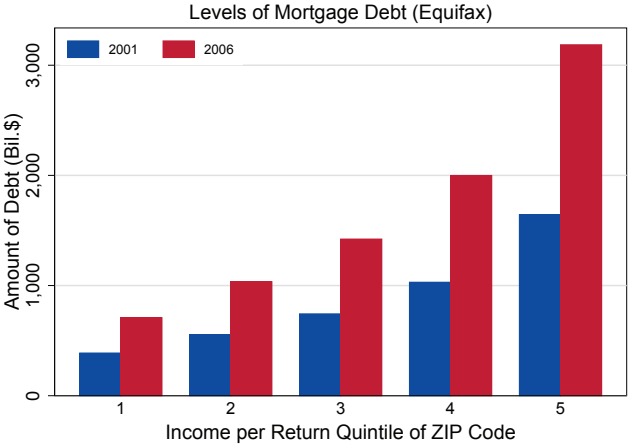




9

- Debt shifts up acoss all income
- Regressions show no change in slope
- Within cities, no change
- Identical results from SCF using individual level data

Finding 2′: Most new debt went to high income households



- Debt highly skewed toward rich people
- In 2006

Quintile	AGI	Riskscore
1	32	656
2	39	689
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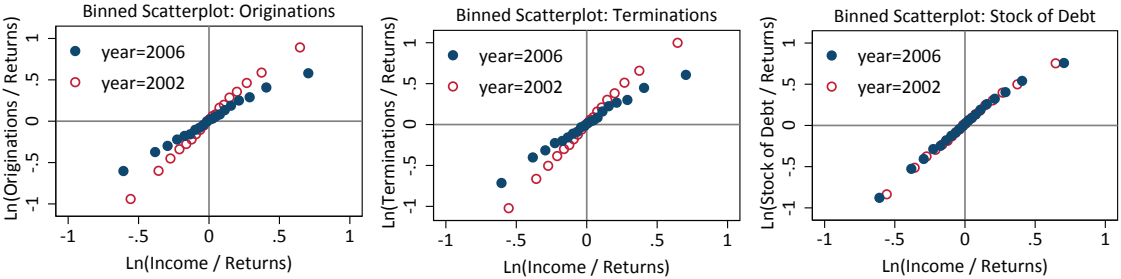
- In $, most debt went to the rich.

Finding 3: Relative changes in originations offset by terminations

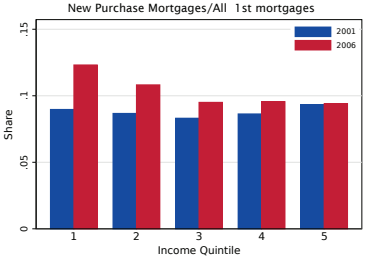


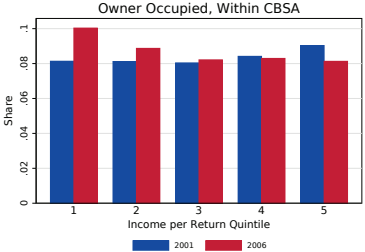
- Growth in stock of debt is

$$\underbrace{\Delta\ \text{Stock}}_{\text{Net flow}} = \underbrace{\text{Purchases} + \text{Cash-Out}}_{\text{Gross flow in}} - \underbrace{\text{Curtailment} - \text{Amortization} - \text{Sales}}_{\text{Gross flow out}}$$

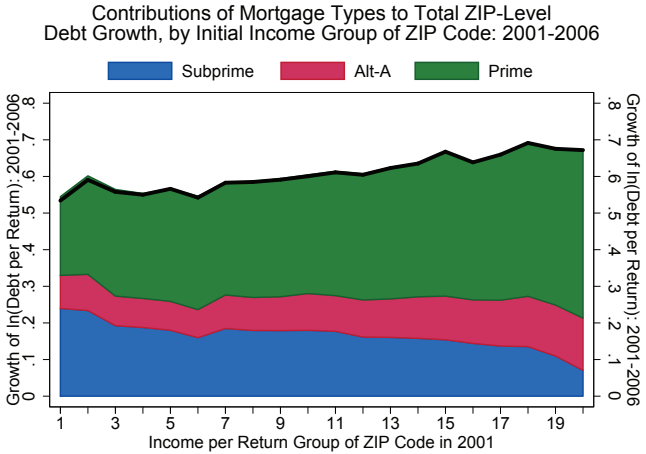
Finding 4: Relative increase in churn

- In marginal areas, there *was* an increase in transactions (Adelino, et al. (2016), Mian and Sufi (2009, 2016)).
- People bought more homes
- People also *sold* more homes
- Role of investors

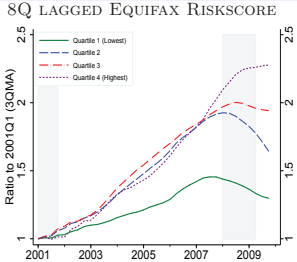


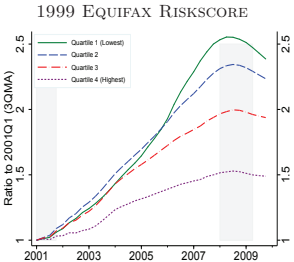


Finding 7: Subprime didn't cause a reallocation, it prevented one



- Subprime was skewed to low income areas.
- Offset a little by Alt-A
- If we only had prime, boom would have resulted in big shift in debt to high income areas.





(4) Albanesi et al. (2016)

- Growth in debt roughly equal across credit scores.
- Problem: Endogeneity of credit scores
  1. Individual: Homeownership ⇒ Higher credit score
  2. Geographic: High HPA ⇒ Higher credit score?

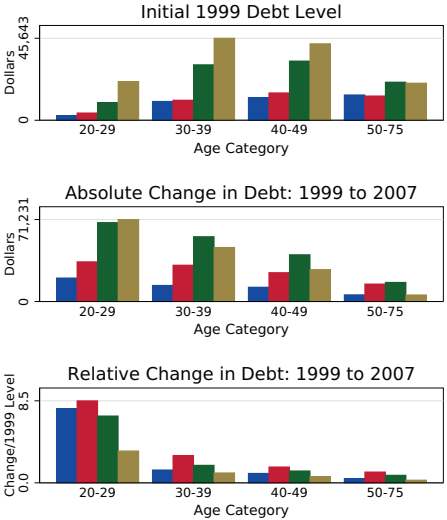
	House price growth in %				
Score quintile	<40%	40-75%	75-105%	105-130%	>130%
1	-	9.9	8.5	13	13.8
2	-	11	13.9	16.4	17.3
3	-	6.6	8.4	11.3	11.3
4	-	3.6	2	4.1	1.9
5	-	-0.8	-2.9	-2.3	-3

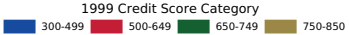
- Mian and Sufi (2016): *use 1999 credit scores.*

	Depvar=P(Default)		
	(1)	(2)	(3)
$Score_{2006}$	-13.423	-11.105	
$Score_{1998}$		-2.318	-13.423
$Score_{2006} - Score_{1998}$	2.318		-11.105

(4') 1999 Credit scores and endogeneity

- Why does debt grow faster for low 1999 credit score versus high?
  1. Age: Credit Scores is *highly* correlated with age
  2. Mean Reversion: Low credit score people have very little debt
- Identification requires "normal" growth in mortgage debt independent
  1. credit score
  2. level of debt
- Identification problem using 1999 credit scores is worse than using 8-quarter lagged credit scores.

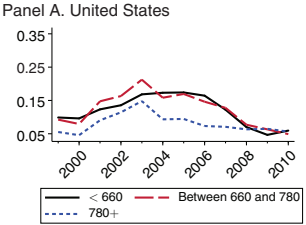




(5) Neil Bhutta

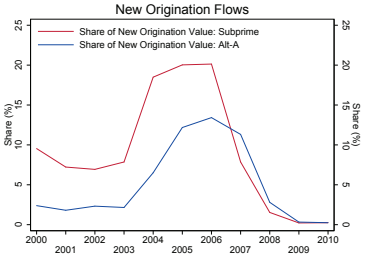
- Credit score dimension of the boom limited
- Avoids endogeneity problem by looking at credit score before credit.
  1. First time home buyers from Bhutta (2015, JMonE) (we follow Bhutta for "extensive margin" above)
  2. Equity extraction from Bhutta and Keys (2016, AER)

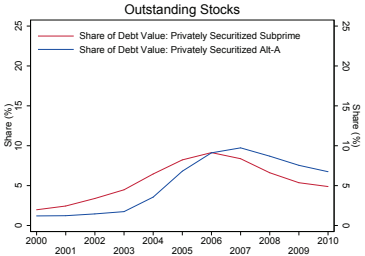
	1999	2001	2003	2005	2007
Initial credit score	A. 20–39 year olds				
All	0.111	0.111	0.108	0.092	0.062
All (has score)	0.121	0.119	0.117	0.101	0.069
740+	0.184	0.219	0.211	0.163	0.138
680-739	0.173	0.190	0.189	0.156	0.112
620-679	0.151	0.150	0.151	0.129	0.083
< 620	0.077	0.063	0.063	0.058	0.032



(7) What about Subprime?

- Much research, including our own showed the explosive growth of subprime in the early 2000s
- How is this consistent with our current research?
  1. Stocks versus flows
  2. Subprime was replacing FHA





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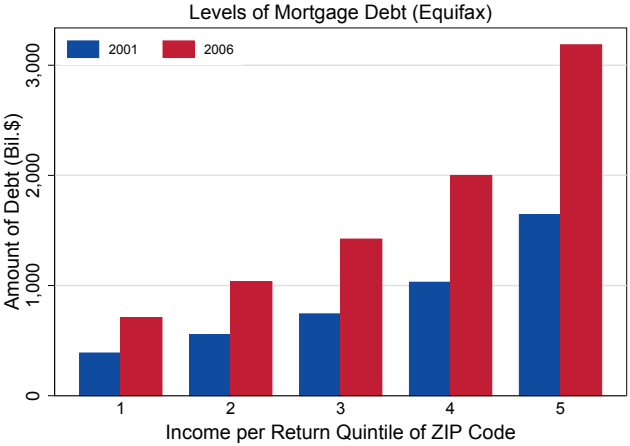

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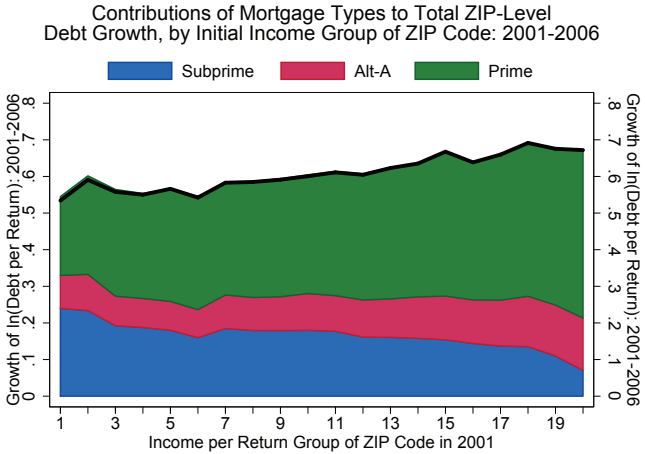

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The slide you've all been waiting for...

- The end.